AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Design Phase – Solution Document

Project Type: ServiceNow IT Helpdesk Automation | Platform: ServiceNow

1. Design Overview

The proposed solution is designed to automate the classification of IT helpdesk incidents in ServiceNow. Users may submit incidents related to Wi-Fi connectivity, projector failures, password problems, slow computers, and other common IT issues. The system reads the incident Short Description and Description, identifies relevant keywords, determines the appropriate category and subcategory, and updates the ticket automatically through ServiceNow Flow Designer.

2. Problem Definition

In a conventional helpdesk process, incidents are manually reviewed and categorized by support staff. This can result in inconsistent categorization, additional processing time, incorrect routing, and delays in assigning tickets to the appropriate support team. The proposed design reduces these manual activities by introducing a rule-based classification flow.

3. Proposed Solution Architecture

Layer	Design Component	Purpose
Input	Incident form / user request	Captures Short Description and Description.
Trigger	Flow Designer – Incident Created/Updated	Starts classification automatically.
Analysis	Keyword / condition evaluation	Identifies issue-related keywords.
Classification	Category & Subcategory rules	Maps the issue to a predefined classification.
Output	Incident record update	Writes Category and Subcategory back to the ticket.

4. Design Objectives

- Automatically classify newly created or updated incidents based on meaningful words in the request.
- Maintain a consistent Category and Subcategory structure across the helpdesk.
- Reduce manual classification effort and improve initial ticket handling.
- Keep the design simple, maintainable, and extendable for additional issue types.
- Provide a clear workflow that can be tested and demonstrated within ServiceNow.

5. High-Level Process

Incident Created → Flow Trigger → Read Incident Text → Check Keywords/Conditions → Select Category → Select Subcategory → Update Incident → End

The design intentionally keeps the classification decision inside Flow Designer so that the logic can be viewed, tested, maintained, and modified by a ServiceNow administrator without changing the incident form manually.

DESIGN PHASE – CLASSIFICATION & FLOW LOGIC

6. Classification Design

The classification engine uses a predefined mapping between common helpdesk keywords and ServiceNow Category/Subcategory values. The rules are applied in a controlled sequence. More specific conditions should be evaluated before broad conditions to reduce the possibility of incorrect classification.

Issue Type	Example Keywords	Category	Subcategory
Wi-Fi / Network	wifi, wi-fi, wireless, network, internet connection	Network	Wi-Fi / Connectivity
Projector	projector, display, projection, HDMI	Hardware	Projector
Password	password, reset password, login password, credentials	Access	Password / Account
Slow Computer	slow, lagging, performance, hangs, freezes	Hardware	Performance
Other / Unclassified	no matching keyword	General	Other

7. Flow Designer Logic

The Flow Designer implementation can be designed using the following logical sequence:

- Trigger: Start when an Incident is created. If required, the same design can also support an Incident update trigger.
- Get Incident Data: Use the triggering Incident record as the input data pill.
- Normalize/Inspect Text: Evaluate Short Description and Description for classification keywords.
- Decision: Use If/Else conditions to compare the incident text against the defined keyword groups.
- Set Classification: Assign the appropriate Category and Subcategory values.
- Update Record: Update the same Incident record with the selected classification.
- Fallback: If no rule matches, keep the incident in a General/Other classification for manual review.

8. Decision Logic Example

Priority	Condition	Action
1	Text contains Wi-Fi/network keywords	Category = Network; Subcategory = Wi-Fi / Connectivity
2	Text contains projector/display keywords	Category = Hardware; Subcategory = Projector
3	Text contains password/login keywords	Category = Access; Subcategory = Password / Account
4	Text contains slow/performance keywords	Category = Hardware; Subcategory = Performance
5	No condition matched	Category = General; Subcategory = Other

9. Data & Field Design

The design primarily uses existing ServiceNow Incident fields. The Short Description and Description fields act as the classification input. Category and Subcategory act as the classification output. If the project requires a custom table, the same design can be extended by storing classification rules separately and referencing them from the flow.

10. Design Considerations

- Use consistent keyword spelling and avoid overlapping rules where possible.
- Place specific rules before generic rules.
- Provide a fallback path for tickets that cannot be classified automatically.
- Keep business logic visible and easy to modify in Flow Designer.

DESIGN PHASE – VALIDATION, SECURITY & EXPECTED RESULT

11. Testing & Validation Design

The solution should be validated with representative incident requests. Each test case verifies that the correct category and subcategory are selected and that the incident record is updated successfully.

Test Case	Sample Request	Expected Result
TC-01	“Wi-Fi is not connecting in the classroom.”	Network → Wi-Fi / Connectivity
TC-02	“Projector is not displaying the laptop screen.”	Hardware → Projector
TC-03	“I forgot my password and cannot log in.”	Access → Password / Account
TC-04	“The computer is very slow and keeps freezing.”	Hardware → Performance
TC-05	“Need help with an issue not listed above.”	General → Other

12. Error Handling & Exception Design

- If the incident text is empty, the flow should not fail; it should route the ticket to the fallback classification.
- If multiple keyword groups are detected, the priority order of the rules determines the selected classification.
- If a Category/Subcategory value is unavailable or invalid, the flow should stop safely and leave the ticket available for manual handling.
- Flow execution should be reviewed in ServiceNow Flow Designer execution details during testing.

13. Security & Access Design

- Only authorized ServiceNow users should be allowed to modify the flow and classification rules.
- The flow should update only the intended Incident fields.
- The design should follow existing ServiceNow roles, ACLs, and data access policies.
- No unnecessary personal information should be copied into custom fields or external systems.

14. Expected Benefits

- Faster initial classification of common IT support incidents.
- Consistent use of Category and Subcategory values.
- Reduced repetitive manual work for helpdesk staff.
- Improved visibility and reporting because incidents are classified systematically.
- Easy future expansion by adding new keyword groups and classification rules.

15. Final Design Summary

The final design provides an automated, rule-based incident classification mechanism using ServiceNow Flow Designer. When an IT helpdesk incident is created, the flow evaluates the request text, matches it against predefined issue patterns, selects the corresponding Category and Subcategory, and updates the incident automatically. A fallback path handles unmatched requests. The design is modular, testable, and suitable for extension as additional helpdesk issue types are introduced.

16. Proposed Implementation Sequence

1. Finalize Category/Subcategory values → 2. Create/update classification rules → 3. Build Flow Designer trigger → 4. Add decision branches → 5. Update Incident record → 6. Test with sample incidents → 7. Validate execution → 8. Demonstrate final automation

Document Status: Design Phase Solution | Project: Auto Ticket Classification using Flow Designer | Platform: ServiceNow